

Faculty of Technology and Engineering Chandubhai S. Patel Institute of Technology (CSPIT) Department of Computer Science & Engineering					
Date: / /					
<u>Laboratory Manual</u>					
Academic Year	:	2026-27	Semester	:	3 rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS
<u>Practical – 5</u>					
<u>PROBLEM STATEMENT :</u> Create a student registration form using HTML5 input types, JavaScript validation, Regular Expressions, password strength indicator, inline error messages, and accessibility attributes. The form must collect Name, Email, Mobile, Password, Confirm Password, Course, Year, Gender, and Terms acceptance.					
<u>OBJECTIVE :</u> To design and implement a fully client-side validated student registration form using HTML5 , CSS3 , and JavaScript . The objective is to understand HTML5 input types and attributes, apply Regular Expressions for field validation, implement a real-time password strength indicator, display contextual error messages adjacent to each field, and create an accessible, keyboard-friendly form following WCAG 2.1 AA guidelines — all without any external JavaScript library.					
25CS001 – Ansh Adodari			ITUE203 : WEB DEVELOPMENT FRAMEWORKS		

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CODE :

1) Registration Form — index.html

HTML Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/validation.js" defer></script>
</head>
<body>
  <main class="page-shell">
    <section class="intro-panel" aria-labelledby="page-title">
      <h1 id="page-title">Student Registration</h1>
      <p class="intro-copy">Create your learner profile by entering accurate details.
    </p>
    </section>

    <section class="form-panel" aria-labelledby="form-title">
      <div class="form-heading">
        <div>
          <p class="section-label">New student profile</p>
          <h2 id="form-title">Personal details</h2>
        </div>
        <span class="required-note">* Required</span>
      </div>

      <form id="registration-form" novalidate>
        <div class="form-grid">

          <div class="field field-wide">
            <label for="full-name">Full name <span>*</span></label>
            <input type="text" id="full-name" name="fullName"
              placeholder="e.g. Aarya Patel"
              autocomplete="name" required>
            <small class="hint">Use 2-50 letters and spaces.</small>
            <p class="error-message" id="full-name-error" aria-live="polite"></p>
          </div>

          <div class="field">
            <label for="email">Email address <span>*</span></label>
            <input type="email" id="email" name="email"
              placeholder="student@example.com"
              autocomplete="email" required>
            <p class="error-message" id="email-error" aria-live="polite"></p>
          </div>
        </div>
      </form>
    </section>
  </main>
</body>
</html>
```

```

<div class="field">
  <label for="mobile">Mobile number <span>*</span></label>
  <input type="tel" id="mobile" name="mobile"
    placeholder="9876543210"
    inputmode="numeric" maxlength="10"
    autocomplete="tel" required>
  <p class="error-message" id="mobile-error" aria-live="polite"></p>
</div>

<div class="field">
  <label for="password">Password <span>*</span></label>
  <input type="password" id="password" name="password"
    placeholder="Create a strong password"
    autocomplete="new-password" required>
  <small class="hint">8+ chars, uppercase, lowercase, number and symbol.
</small>

  <div id="strength-bar-wrap" class="strength-bar-wrap">
    <div class="strength-seg" id="seg1"></div>
    <div class="strength-seg" id="seg2"></div>
    <div class="strength-seg" id="seg3"></div>
    <div class="strength-seg" id="seg4"></div>
  </div>
  <small id="strength-label" class="strength-label"></small>
  <p class="error-message" id="password-error" aria-live="polite"></p>
</div>

<div class="field">
  <label for="confirm-password">Confirm password <span>*</span></label>
  <input type="password" id="confirm-password" name="confirmPassword"
    placeholder="Re-enter your password"
    autocomplete="new-password" required>
  <p class="error-message" id="confirm-password-error" aria-live="polite"></p>
</div>

<div class="field">
  <label for="course">Course <span>*</span></label>
  <select id="course" name="course" required>
    <option value="">Select a course</option>
    <option value="BCA">BCA - Computer Applications</option>
    <option value="BBA">BBA - Business Administration</option>
    <option value="BSC">B.Sc. - Information Technology</option>
    <option value="BTECH">B.Tech - Computer Engineering</option>
  </select>
  <p class="error-message" id="course-error" aria-live="polite"></p>
</div>

<div class="field">
  <label for="year">Year of study <span>*</span></label>
  <select id="year" name="year" required>
    <option value="">Select year</option>
    <option value="1">First year</option>
    <option value="2">Second year</option>
    <option value="3">Third year</option>
    <option value="4">Fourth year</option>
  </select>
  <p class="error-message" id="year-error" aria-live="polite"></p>
</div>
</div>

```

```
<fieldset class="radio-fieldset">
  <legend>Gender <span>*</span></legend>
  <div class="radio-options">
    <label><input type="radio" name="gender" value="Female"> Female</label>
    <label><input type="radio" name="gender" value="Male"> Male</label>
    <label><input type="radio" name="gender" value="Other"> Other</label>
  </div>
  <p class="error-message" id="gender-error" aria-live="polite"></p>
</fieldset>

<label class="terms-label">
  <input type="checkbox" id="terms" name="terms">
  <span>I confirm the information is correct and accept the
    terms of registration. <strong>*</strong></span>
</label>
<p class="error-message" id="terms-error" aria-live="polite"></p>

<div class="form-actions">
  <button type="reset" class="secondary-button">Clear form</button>
  <button type="submit" class="primary-button">Register student →</button>
</div>
<p id="form-status" class="form-status" role="status" aria-live="polite"></p>
</form>
</section>

<section id="success-panel" class="success-panel" hidden>
  <p class="success-mark" aria-hidden="true">✓</p>
  <div>
    <h2>Welcome, <span id="registered-name"></span>.</h2>
    <p>Your <span id="registered-course"></span> profile is ready
      for the <span id="registered-year"></span>.</p>
  </div>
</section>
</main>
</body>
</html>
```

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2) JavaScript Validation & Password Strength — js/validation.js

Javascript Code:

```
(function () {  
    'use strict';  
  
    var form          = document.getElementById('registration-form');  
    var successPanel = document.getElementById('success-panel');  
  
    /* — Regular Expression patterns — */  
    var patterns = {  
        name:      /^[A-Za-z] [A-Za-z ]{1,49}$/,  
        email:     /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/,  
        mobile:    /^[6-9][0-9]{9}$/,  
        password: /^(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9])(?=.*[^\A-Za-z0-9]).{8,}$/  
    };  
  
    /* — Error helpers — */  
    function setError(field, message) {  
        var err = document.getElementById(field.id + '-error');  
        field.setAttribute('aria-invalid', 'true');  
        err.textContent = message;  
        return false;  
    }  
  
    function clearError(field) {  
        var err = document.getElementById(field.id + '-error');  
        field.removeAttribute('aria-invalid');  
        if (err) err.textContent = '';  
    }  
  
    /* — Password strength — */  
    var pwInput  = document.getElementById('password');  
    var segs     = [1,2,3,4].map(function(n){ return document.getElementById('seg'+n); });  
    var strLabel = document.getElementById('strength-label');  
    var COLORS   = ['', '#e53e3e', '#ed8936', '#48bb78', '#2f855a', '#2f855a'];  
    var LABELS   = ['', 'Weak', 'Fair', 'Good', 'Strong', 'Strong'];  
  
    function getScore(pw) {  
        var s = 0;  
        if (pw.length >= 8)          s++;  
        if (/[a-z]/.test(pw))        s++;  
        if (/[A-Z]/.test(pw))        s++;  
        if (/[0-9]/.test(pw))        s++;  
        if (/^[^\A-Za-z0-9]/.test(pw)) s++;  
        return s;  
    }  
  
    if (pwInput) {  
        pwInput.addEventListener('input', function () {  
            var score = getScore(this.value);
```

```

        var col    = this.value.length ? COLORS[score] : '';
        segs.forEach(function(seg, i) {
            seg.style.background = (i < score && col) ? col : '#e2e8f0';
        });
        strLabel.textContent = this.value.length ? LABELS[score] : '';
        strLabel.style.color = col;
    });
}

/* — Full form validation — */
function validateForm() {
    var valid      = true;
    var fullName   = document.getElementById('full-name');
    var email      = document.getElementById('email');
    var mobile     = document.getElementById('mobile');
    var password   = document.getElementById('password');
    var confirm    = document.getElementById('confirm-password');
    var course     = document.getElementById('course');
    var year       = document.getElementById('year');
    var terms      = document.getElementById('terms');
    var gender     = document.querySelector('input[name="gender"]:checked');

    [fullName, email, mobile, password, confirm, course, year].forEach(clearError);
    document.getElementById('gender-error').textContent = '';
    document.getElementById('terms-error').textContent = '';

    if (!patterns.name.test(fullName.value.trim()))
        valid = setError(fullName, 'Enter a name with 2-50 letters and spaces.') && valid;
    if (!patterns.email.test(email.value.trim()))
        valid = setError(email, 'Enter a valid email address.') && valid;
    if (!patterns.mobile.test(mobile.value.trim()))
        valid = setError(mobile, 'Enter a 10-digit number starting with 6, 7, 8 or 9.') &&
valid;
    if (!patterns.password.test(password.value))
        valid = setError(password, 'Use 8+ characters with uppercase, lowercase, number
and symbol.') && valid;
    if (confirm.value !== password.value || confirm.value === '')
        valid = setError(confirm, 'Passwords must match.') && valid;
    if (course.value === '')
        valid = setError(course, 'Select a course.') && valid;
    if (year.value === '')
        valid = setError(year, 'Select your year of study.') && valid;
    if (!gender) {
        document.getElementById('gender-error').textContent = 'Select a gender option.';
        valid = false;
    }
    if (!terms.checked) {
        document.getElementById('terms-error').textContent = 'Accept the terms to
continue.';
        valid = false;
    }
    return valid;
}

/* — Submit handler — */
form.addEventListener('submit', function (e) {
    e.preventDefault();
    var status = document.getElementById('form-status');
    if (!validateForm()) {

```

```
        status.textContent = 'Please correct the highlighted fields.';
        status.className = 'form-status error';
        var first = form.querySelector('[aria-invalid="true"]');
        if (first) first.focus();
        return;
    }
    status.textContent = '';
    var sel = document.getElementById('course');
    document.getElementById('registered-name').textContent
        = document.getElementById('full-name').value.trim();
    document.getElementById('registered-course').textContent
        = sel.options[sel.selectedIndex].text;
    var yr = document.getElementById('year');
    document.getElementById('registered-year').textContent
        = yr.options[yr.selectedIndex].text.toLowerCase();
    successPanel.hidden = false;
    successPanel.scrollIntoView({ behavior: 'smooth', block: 'center' });
});

/* — Reset handler — */
form.addEventListener('reset', function () {
    window.setTimeout(function () {
        form.querySelectorAll('input, select').forEach(clearError);
        ['gender-error', 'terms-error', 'form-status'].forEach(function(id) {
            document.getElementById(id).textContent = '';
        });
        segs.forEach(function(s) { s.style.background = '#e2e8f0'; });
        if (strLabel) strLabel.textContent = '';
        successPanel.hidden = true;
    }, 0);
});

}());
```

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3) Styles — css/style.css

CSS Code:

```
@import url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;700&family=Space+Grotesk:wght@500;600;700&display=swap');

:root {
  --ink:          #17212b;
  --muted:        #66737f;
  --line:         #d8e0e5;
  --accent:       #e87532;
  --accent-dark:  #b95019;
  --teal:         #146c68;
  --danger:       #b42318;
  --success:      #116149;
}

body { margin:0; font-family:'DM Sans',sans-serif;
       background:#dfeae5; color:var(--ink); line-height:1.5; }

/* - Layout - */
.page-shell { max-width:1080px; margin:0 auto; padding:48px 22px 64px; }
.form-panel { padding:clamp(22px,5vw,46px); background:#fff;
              box-shadow:0 18px 45px rgba(23,33,43,.1); }
.form-grid  { display:grid; grid-template-columns:repeat(2,1fr);
              gap:20px; margin-top:25px; }
.field-wide { grid-column:span 2; }

/* - Inputs - */
input, select {
  width:100%; min-height:46px; padding:11px 13px;
  border:1px solid var(--line); border-radius:2px;
  font:inherit; outline:none;
  transition:border-color .2s, box-shadow .2s;
}
input:focus, select:focus {
  border-color:var(--teal);
  box-shadow:0 0 0 3px rgba(20,108,104,.13);
}
input[aria-invalid="true"], select[aria-invalid="true"] {
  border-color:var(--danger);
}

/* - Error messages - */
.error-message { min-height:18px; margin:4px 0 0;
                color:var(--danger); font-size:.75rem; }

/* - Password strength bar - */
.strength-bar-wrap { display:flex; gap:4px; margin:6px 0 2px; }
.strength-seg      { height:5px; border-radius:3px; flex:1;
```

```
        background:#e2e8f0; transition:background .25s; }
.strength-label { font-size:.75rem; font-weight:700; }

/* - Radio and Checkbox - */
.radio-fieldset { margin:12px 0 0; padding:0; border:0; }
.radio-options { display:flex; flex-wrap:wrap; gap:10px 28px; margin-top:10px; }
.terms-label { display:flex; align-items:flex-start; gap:10px; margin-top:14px; }
input[type="radio"], input[type="checkbox"] {
  width:17px; min-height:17px; accent-color:var(--teal);
}

/* - Buttons - */
button { min-height:44px; padding:10px 18px; border:0; border-radius:2px;
  font:inherit; font-weight:700; cursor:pointer; }
.primary-button { color:#fff; background:var(--accent); }
.primary-button:hover { background:var(--accent-dark); }
.secondary-button { border:1px solid var(--line); background:transparent; }

/* - Success panel - */
.success-panel { display:flex; align-items:center; gap:18px;
  margin-top:24px; padding:20px 24px;
  background:#fff; border:1px solid var(--line); }
.success-mark { display:grid; place-items:center;
  width:40px; height:40px; border-radius:50%;
  background:var(--success); color:#fff;
  font-size:1.35rem; font-weight:700; }

/* - Responsive - */
@media (max-width:640px) {
  .form-grid { grid-template-columns:1fr; }
  .field-wide { grid-column:auto; }
}
```

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OUTPUT :**Screenshot 1 – Complete Registration Form (Empty State)**

- All 9 fields visible in 2-column grid layout
- Full Name (wide), Email, Mobile, Password + strength bar, Confirm Password
- Course & Year dropdowns, Gender radio buttons, Terms checkbox
- Clear form and Register student buttons

*Fig 1: Complete registration form (empty state)***Screenshot 2 – Valid Input + Successful Registration**

- All fields correctly filled (Name: Aarya Patel)
- Email: aarya@example.com, Mobile: 9876543210
- Course: B.Tech CE, Year: First year, Gender: Female
- Success panel: "Welcome, Aarya Patel."

*Fig 2: Valid submission — success panel displayed***Screenshot 3 – Invalid Name / Email / Mobile**

- Name "A1" — error: "Enter a name with 2-50 letters and spaces."
- Email "invalid-email" — error: "Enter a valid email address."
- Mobile "12345" — error: "Enter a 10-digit number starting with 6, 7, 8 or 9."

*Fig 3: Invalid name, email & mobile validation errors***Screenshot 4 – Password & Confirm Password Validation**

- Password too weak — error: "Use 8+ characters with uppercase, lowercase, number and symbol."
- Confirm password mismatch — error: "Passwords must match."
- Both fields highlighted with red border

*Fig 4: Password & confirm-password validation errors***Screenshot 5 – Password Strength Indicator**

- Real-time 4-segment strength bar below password field
- Strong password — all 4 segments filled, label: "Strong"
- Weak password — 1 red segment, label: "Weak"

*Fig 5: Real-time password strength bar***Screenshot 6 – Terms Not Accepted**

- All other fields valid — no other errors shown
- Terms checkbox unchecked
- Single error: "Accept the terms to continue."

*Fig 6: Terms checkbox not accepted — single error***25CS001 – Ansh Adodari****ITUE203 : WEB DEVELOPMENT FRAMEWORKS**

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GITHUB LINK :

https://github.com/25cs001ansh/Web_Technology.git

LEARNING OUTCOME :

- ✓ Understood how to use correct **HTML5 input types** (`type="email"`, `type="tel"`, `type="password"`, `type="radio"`, `type="checkbox"`) along with attributes such as `autocomplete`, `inputmode`, `maxlength`, and `novalidate` to build a semantically rich registration form.
- ✓ Learned to write **Regular Expressions** for four fields — name (`/^[A-Za-z][A-Za-z ]{1,49}$/`), email, 10-digit Indian mobile number (`/^[6-9][0-9]{9}$/`), and a strong-password pattern using **lookaheads** — and apply them via `RegExp.test()` in JavaScript.
- ✓ Implemented a complete **client-side validation** function inside an IIFE using `'use strict'`, with helper functions `setError()` and `clearError()` to attach/remove `aria-invalid` and display all errors simultaneously on form submission.
- ✓ Built a **real-time password strength meter** using a five-criteria scoring algorithm (`length ≥ 8`, lowercase, uppercase, digit, symbol) and mapped scores to four coloured bar segments updated on every `input` event.
- ✓ Practiced placing **inline error messages** adjacent to each field using dedicated `<p class="error-message">` elements, giving users immediate, field-level feedback that is clear and actionable.
- ✓ Applied **ARIA accessibility** attributes — `aria-live="polite"` on error containers, `aria-invalid="true"` on invalid fields, `<fieldset>` + `<legend>` for the radio group — and programmatic `focus()` on the first invalid field to ensure full keyboard and screen-reader compatibility (WCAG 2.1 AA).
- ✓ Understood the principle of **separation of concerns** — HTML defines structure, CSS handles all visual styling and transitions, and JavaScript only toggles state — keeping the codebase clean and maintainable without any external library.

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